

CRANBROOK, BC, Canada

WMO: 716600

Lat: 49.6175N

Lon: 115.7892W

Elev: 3050

StdP: 13.15

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-8.4	-2.7	-16.8	2.5	-6.2	-10.3	3.5	-2.2	17.8	34.7	16.1	35.1	1.8	0	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	89.2	61.3	85.9	60.2	82.4	59.2	63.4	82.6	61.8	80.7	60.3	78.7	8.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.0	77.7	66.9	54.9	71.9	66.2	52.9	66.7	65.6	30.3	82.8	29.1	80.8	28.0	79.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.6	13.6	-16.3	94.3	7.2	3.6	-21.6	96.9	-25.8	99.1	-29.9	101.1	-35.1	103.8		
(5)			-16.8	66.5	7.1	2.6	-22.0	68.4	-26.1	69.9	-30.1	71.3	-35.3	73.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	21.7	25.5	35.2	43.5	52.5	58.8	67.1	65.6	55.7	42.0	30.2	20.5	
	DBStd	18.08	11.85	10.23	8.06	6.34	6.95	6.28	6.22	5.98	7.13	7.29	9.52	10.86	
	HDD50	4092	878	687	460	212	52	5	0	0	25	263	595	915	
	HDD65	8156	1343	1107	924	645	388	206	51	67	287	713	1045	1380	
	CDD50	1644	0	0	1	17	131	268	531	484	197	15	0	0	
	CDD65	233	0	0	0	0	2	19	117	86	9	0	0	0	
	CDH74	3600	0	0	0	2	105	333	1641	1289	230	0	0	0	
	CDH80	1246	0	0	0	0	12	87	635	459	53	0	0	0	
Wind	WSAvg	5.0	3.3	4.0	5.6	6.1	6.2	5.9	6.1	5.7	5.1	4.8	3.9	3.2	
Precipitation	PrecAvg	16.1	1.4	0.9	0.8	1.1	1.8	2.0	1.3	1.4	1.3	0.7	1.4	1.6	
	PrecMax	21.3	3.4	2.4	1.3	2.5	4.7	6.1	3.7	3.4	3.9	1.8	3.6	4.0	
	PrecMin	11.5	0.2	0.2	0.2	0.1	0.4	0.3	0.2	0.0	0.0	0.1	0.2	0.4	
	PrecStd	2.4	0.9	0.6	0.3	0.7	1.1	1.3	0.7	0.9	1.0	0.5	0.9	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	48.7	60.3	72.6	81.5	88.0	94.1	92.6	85.6	68.9	53.4	43.5	
		MCWB	39.4	40.4	44.8	52.1	56.8	61.0	62.7	61.5	58.9	54.1	46.1	38.4	
	2%	DB	41.7	43.9	54.4	66.4	77.0	81.7	90.3	88.6	80.5	62.5	48.6	40.0	
		MCWB	37.0	37.1	42.3	49.3	55.1	58.8	61.9	60.8	57.8	50.3	43.4	36.3	
	5%	DB	39.0	41.0	50.3	61.1	72.3	77.5	87.0	85.4	75.6	57.9	45.1	37.4	
		MCWB	35.2	35.5	40.5	46.4	53.3	57.1	61.1	59.6	56.2	47.8	40.4	34.4	
	10%	DB	36.3	38.2	46.8	56.5	67.4	73.1	83.3	81.7	70.9	53.9	42.2	34.7	
		MCWB	33.3	33.6	38.6	44.0	51.2	55.7	60.0	58.8	54.4	45.5	38.4	32.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	41.4	46.6	53.2	58.7	63.4	66.3	64.6	61.5	55.3	47.8	39.9
			MCDB	44.1	46.2	55.6	69.0	77.1	83.2	87.3	83.9	78.3	66.8	52.3	42.3
2%		WB	37.5	38.2	43.6	50.0	56.4	60.5	64.1	62.6	58.9	51.6	43.9	36.8	
		MCDB	40.6	42.6	52.4	64.4	72.6	77.5	83.1	82.4	77.0	60.4	48.0	39.4	
5%		WB	35.6	35.9	41.4	47.2	54.4	58.5	62.6	61.1	57.0	48.8	41.1	34.8	
		MCDB	38.7	40.1	48.7	59.2	69.3	74.3	81.6	80.5	73.4	56.6	44.2	37.3	
10%		WB	33.4	34.0	39.4	44.9	52.3	56.6	61.0	59.7	55.1	46.2	38.8	32.3	
		MCDB	36.1	37.7	45.9	55.0	65.5	70.2	79.7	78.2	69.6	52.8	41.7	34.5	
Mean Daily Temperature Range	5% DB	MDBR	13.1	16.8	17.8	20.9	22.4	22.1	26.4	27.3	24.4	19.1	13.0	11.8	
		MCDBR	13.8	17.0	23.1	29.9	30.5	29.4	32.1	33.2	32.8	24.8	15.5	12.6	
		MCWBR	10.3	11.6	13.5	15.8	14.3	12.7	12.0	12.8	14.7	14.0	11.0	9.8	
	5% WB	MCDBR	13.0	15.5	20.9	26.8	27.0	26.2	28.6	29.5	30.2	21.9	14.5	12.0	
		MCWBR	9.8	11.1	12.8	14.6	13.4	12.0	11.6	12.2	14.0	12.9	10.8	9.6	
Clear-Sky Solar Irradiance	taub	0.256	0.263	0.286	0.351	0.361	0.361	0.358	0.370	0.336	0.298	0.273	0.249		
	taud	2.358	2.349	2.370	2.264	2.354	2.403	2.423	2.397	2.472	2.565	2.514	2.393		
	Ebn at Noon	257	284	295	283	284	285	284	275	274	270	252	245		
	Edh at Noon	20	26	31	39	37	36	35	34	28	22	18	17		
All-Sky Solar Radiation	RadAvg	333	635	951	1401	1674	1791	2110	1747	1214	716	350	260		
	RadStd	35	58	101	117	143	152	170	133	131	84	39	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page